Q1. Big Data means a large amount of data that normal tools cannot process.
It requires special technologies to store and analyze it.
It includes different types of data like text, images, and videos.

Q2. MapReduce is used in Hadoop to process big data.
It divides the work into map and reduce steps.
Then it combines the results.

Q3. The 5 V's are Volume, Velocity, Variety, Veracity, and Value.
They describe characteristics of Big Data.
They explain size, speed, type, accuracy, and importance.

Q4. There are three types of Big Data: structured, semi-structured, and unstructured.
Structured data is organized, and unstructured is not.
Semi-structured has partial structure.

Q5Hadoop is a tool used to store and process Big Data.
It works on many computers together.
It uses HDFS and MapReduce.

Q6. **Introduction:**
HDFS is a file system used in Hadoop to store large amounts of data across many computers. It is designed for Big Data applications.

Architecture:
HDFS has two main parts:

- NameNode → manages the system
- DataNodes → store the actual data

The NameNode keeps track of file information and where data is stored. DataNodes store the data in the form of blocks.

Files are divided into blocks and distributed across different machines. This helps in storing large files efficiently.

How it works:
When a user wants to access a file, the request goes to the NameNode first. Then the client directly interacts with DataNodes to read or write data.

Important Features:

- Data is stored in blocks
- Multiple copies are stored (replication)
- System is fault-tolerant

DataNodes send heartbeat signals to NameNode. If a node fails, the system identifies it and handles the failure.

Conclusion:
HDFS architecture helps in storing and processing large data efficiently and safely.

Q7. **Introduction:**
Replication means storing multiple copies of data in HDFS. It is very important for data safety.

Importance:
In HDFS, each block is stored 3 times by default.
This helps in:

- Protecting data from failure
- Providing backup copies
- Improving access speed

When DataNode Fails:
DataNodes send heartbeat signals to NameNode.
If a node fails, heartbeat stops.

Then:

- NameNode detects failure
- It creates new copies of data
- Stores them in other nodes

Users can still access data from other copies.

Conclusion:
Replication ensures that data is safe and always available.